Drew University - CSCI 260 - Fall 2024
In-Class Exercise 3 - LC3 Programming

Instruction: Please work in groups of 3 and write down your names for attendance. 10/16/2022
Names:

Implement the following programs with the LC-3 assembly language in a text editor and load/run it on the simulator website: https://wchargin.com/lc3web/.

Extra Credits: If you submit your programs on Moodle by 10/18 (F) 11:59pm, you will be able to earn extra credits that make up the points lost in HWs.

Note: For both programs, you are **asked** to use LDI and STI to load from and store to the designated data address.

1. (2 points) Write an LC-3 program (starts at x3000) in hexadecimal that **rotates the bit pattern** stored at x4000 to the right by the amount specified by a number stored at x4001. The rotated pattern is then stored at x4002.

   For example, if we set Mem[x4000] = 0001001101110000 and Mem[x4001] = 9, then after the program executes, Mem[x4002] = 1011100000001001.

2. (2 points) Write an LC-3 program (starts at x3000) in hexadecimal that **reverses the bit pattern** stored at x4000. The reversed pattern is then stored at x4001.

   For example, if we set Mem[x4000] = x9871 = 1001 1000 0111 0001, then after the program executes, Mem[x4001] = 1000 1110 0001 1001 = x8E19.